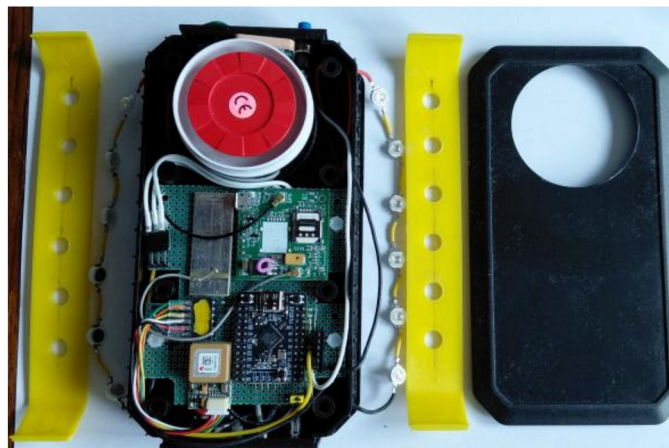
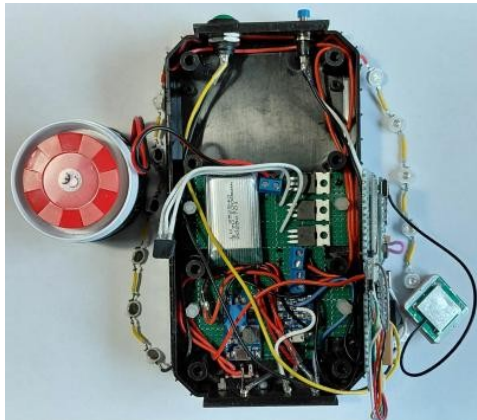


Personal Alarm with GPS and SMS

Warning: The volume of the buzzer may cause hearing damage.



Built with

SIM A7670E	4G successor to the well-known 2G SIM800 modules with a built-in GNSS receiver. The "E" after 7670 indicates the continent. For countries other than Europe, please consult the manual.
SAM M-10Q	Ublox GNSS module
STM32G431CB	STM32 board with compact dimensions

Activate alarm

Press the alarm button

12 x 1 Watt red LEDs flash at a frequency of 25 Hz

Extremely powerful alarm signal (120dB)

The buzzer is disabled in the program using "/" (line 145). To avoid hearing damage, it is recommended to only activate it once everything has been tested and is ready for use.

SMS to a maximum of 5 mobile numbers with a pre-programmed text followed by the current location

Test

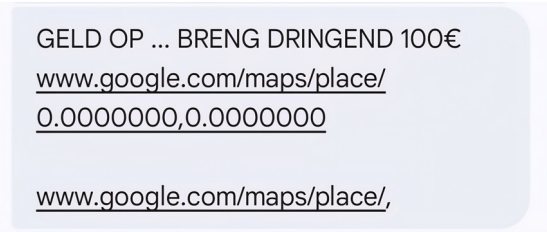
First press the test button, followed by the alarm button.
Then release the alarm button first, and then the test button.

A text message is sent to the number in position 6.

The LEDs light for 1 second.

The buzzer activates for 1 second.

example of sms

A screenshot of an SMS message displayed on a light blue background. The text is black and consists of three lines: 'GELD OP ... BRENG DRINGEND 100€', a URL 'www.google.com/maps/place/0.0000000,0.0000000', and another URL 'www.google.com/maps/place/'.

GELD OP ... BRENG DRINGEND 100€
[www.google.com/maps/place/
0.0000000,0.0000000](http://www.google.com/maps/place/0.0000000,0.0000000)
www.google.com/maps/place/

Usage

Fully programmable by sending an SMS to the module.

The module has 7 programmable positions.

All positions have a maximum length of 50 characters.

The module can only be programmed if PRGM_EN (PA1) is FALSE.

After programming disconnect PA1 from GND

Positions 1 through 5

Enter here the number of the contacts that should be notified with an SMS

Send an SMS in the format

#1 +32123456789#

SMS starts and ends with a #

After the first #, enter the number of the position to be programmed. If it is already programmed, the contents will be erased and overwritten.

Blank

Contact number in international format

SMS ends with a #

Response module via SMS to the number that sent the SMS

pos : 1 tel : +32123456789

Position 6

Enter your mobile phone number here.

During testing, a text message will be sent to this number.

#6 +32987654321#

Response module via SMS to the number that sent the SMS

pos : 6 tel : +32987654321

Position 7

Enter the pre-programmed text here that will be sent via SMS

For teenagers in dire need of money, the message might look like this:

#7 NO MONEY...BRING 100€ URGENTLY#

Response module via SMS to the number that sent the SMS

pos : 7 txt : NO MONEY...BRING 100€ URGENTLY

#LIST#

module responds with

pos : 1 tel: +32123456789

pos : 2 tel: +32123456789

pos : 3 tel: +32123456789

...

pos : 7 txt: NO MONEY...BRING 100€ URGENTLY

#5 DEL#

removes position 5

Answer module

pos 5 : deleted

Parts

1x SIM A7670E module with built-in GNSS.
GPS, Galileo, and GLONASS are supported.

https://nl.aliexpress.com/item/1005004176439432.html?spm=a2g0o.productlist.main.4.565dedfep9sr98&algo_pvid=b9554506-4e30-444e-8212-a62debe6df23&algo_exp_id=b9554506-4e30-444e-8212-a62debe6df23-3&pdp_ext_f=%7B%22order%22%3A%22266%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%2118.15%2112.03%21%21%2120.74%2113.75%21%402103985c17607779760406391e686b%2112000028288768404%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3Ae0abae82%3Bm03_new_user%3A-29895%3BpisId%3A5000000187456134&curPageLogUid=Km9d4QSTdL2y&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005004176439432%7C_p_origin_prod%3A

1x STM32G431CB
https://nl.aliexpress.com/item/1005009686573319.html?spm=a2g0o.productlist.main.3.67cc7ff5zUBYH9&algo_pvid=73fd160f-2a7d-4fd4-ba70-679e92a11b42&algo_exp_id=73fd160f-2a7d-4fd4-ba70-679e92a11b42-2&pdp_ext_f=%7B%22order%22%3A%226%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%212.91%212.91%21%21%2123.68%2123.68%21%4021038df617607780292413985e73ef%2112000049859996012%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3Ae0abae82%3Bm03_new_user%3A-29895&curPageLogUid=F6cO9rzfSHw4&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005009686573319%7C_p_origin_prod%3A

1x SAM M10Q GNSS module
<https://gnss.store/sam-m10q-max-m8q-sam-m8q-gnss-modules/344-elt0169.html>

1x 6 pin JST SH Wire assembly
<https://gnss.store/cable-connectors/13-ELT0002.html>

1x buzzer 12V 120dB
https://nl.aliexpress.com/item/1005008349041301.html?spm=a2g0o.productlist.main.7.c2d465ddhJrOp5&algo_pvid=e5795f61-a1a3-436c-b245-489e4744fc0d&algo_exp_id=e5795f61-a1a3-436c-b245-489e4744fc0d-6&pdp_ext_f=%7B%22order%22%3A%22172%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%211.43%210.86%21%21%2111.60%216.96%21%40211b80e117607773543887046ecb95%2112000044688981331%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3Ae0abae82%3Bm03_new_user%3A-29895&curPageLogUid=ADpHK7lINVYsK&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005008349041301%7C_p_origin_prod%3A

12x LED Rood 1W

https://nl.aliexpress.com/item/1005004932593994.html?spm=a2g0o.productlist.main.9.4ce237772dtxQL&algo_pvid=09025be2-d3ba-47ce-be81-aacf6e3de402&algo_exp_id=09025be2-d3ba-47ce-be81-aacf6e3de402-8&pdp_ext_f=%7B%22order%22%3A%22599%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%213.08%210.87%21%21%213.52%210.99%21%4021039a5b17607781590816208e3431%2112000031065905194%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3Ae0abae82%3Bm03_new_user%3A-29895%3BpisId%3A5000000187456134&curPageLogUid=szkzxegeEe1vn&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005004932593994%7C_p_origin_prod%3A

1x LiPo 1S 1000 mAh 25C

https://nl.aliexpress.com/item/1005006546229827.html?spm=a2g0o.productlist.main.2.1a96kOcEkOcEK6&algo_pvid=e3fbfd8a-1f33-4d3e-a592-940920618da2&algo_exp_id=e3fbfd8a-1f33-4d3e-a592-940920618da2-1&pdp_ext_f=%7B%22order%22%3A%2230%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%217.14%215.00%21%21%218.16%215.71%21%40211b876e17608662385416079ed8f8%2112000037623743634%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A45ea3fc5%3Bm03_new_user%3A-29895&curPageLogUid=JaDfClPZoqwr&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005006546229827%7C_p_origin_prod%3A

of

1x LiPo 1S 1600mAh 25C

https://nl.aliexpress.com/item/1005002695549879.html?spm=a2g0o.store_pc_allItems_or_groupList.0.0.52913fffWa49s0&pdp_npi=4%40dis%21EUR%21%E2%82%AC%207%2C64%21%E2%82%AC%205%2C35%21%21%218.69%216.08%21%40211b612517610294933953476eaabc%2112000021749617747%21sh%21BE%210%21X&gatewayAdapt=glo2nld

1x TP4056 LiPo lader

https://nl.aliexpress.com/item/1005006043031985.html?spm=a2g0o.productlist.main.1.4fdfLuT7LuT7h1&algo_pvid=68174fc6-a4c2-41a1-9fa5-68b88dc87450&algo_exp_id=68174fc6-a4c2-41a1-9fa5-68b88dc87450-0&pdp_ext_f=%7B%22order%22%3A%2210272%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%211.98%210.87%21%21%2116.08%217.05%21%40211b81a317608666438748170e3a5f%2112000035475416223%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A45ea3fc5%3Bm03_new_user%3A-29895%3BpisId%3A5000000187638227&curPageLogUid=evOQqdITMBAO&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005006043031985%7C_p_origin_prod%3A

1x MT3608 DC-DC step up

[https://nl.aliexpress.com/item/1005010130232827.html?
spm=a2g0o.productlist.main.3.6a8d615bBGld3H&algo_pvid=414b53cf-eb7d-440f-9abd-
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url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id
%3A1005010130232827%7C_p_origin_prod%3A](https://nl.aliexpress.com/item/1005010130232827.html?spm=a2g0o.productlist.main.3.6a8d615bBGld3H&algo_pvid=414b53cf-eb7d-440f-9abd-a06dfe36f5a1&algo_exp_id=414b53cf-eb7d-440f-9abd-a06dfe36f5a1-2&pdp_ext_f=%7B%22order%22%3A%229%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%213.04%211.52%21%21%2124.74%2112.37%21%402103894417608667241592453e484a%2112000051246513861%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A45ea3fc5%3Bm03_new_user%3A-29895&curPageLogUid=eZOe8MdoBIbm&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005010130232827%7C_p_origin_prod%3A)

1x housing 171x95x33

[https://nl.aliexpress.com/item/1005005873812461.html?
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%3A1005005873812461%7C_p_origin_prod%3A](https://nl.aliexpress.com/item/1005005873812461.html?spm=a2g0o.productlist.main.43.6d212bd94Btzml&algo_pvid=4e234e03-0412-482b-8f30-3d6a1bcbcb624&algo_exp_id=4e234e03-0412-482b-8f30-3d6a1bcbcb624-42&pdp_ext_f=%7B%22order%22%3A%2243%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%217.22%217.22%21%21%218.19%218.19%21%40211b628117614774656755641e5ac6%2112000034658008044%21sea%21BE%210%21ABX%211%210%21n_tag%3A-29910%3Bd%3A5077d4bf%3Bm03_new_user%3A-29895&curPageLogUid=4cgNG30jmDj9&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005005873812461%7C_p_origin_prod%3A)

3x IRL 540N

3x R 10K

3x R 4K7

1x ON/OFF switch

2x Drukknop N.O.

!!! SOME IMPORTANT NOTES !!!

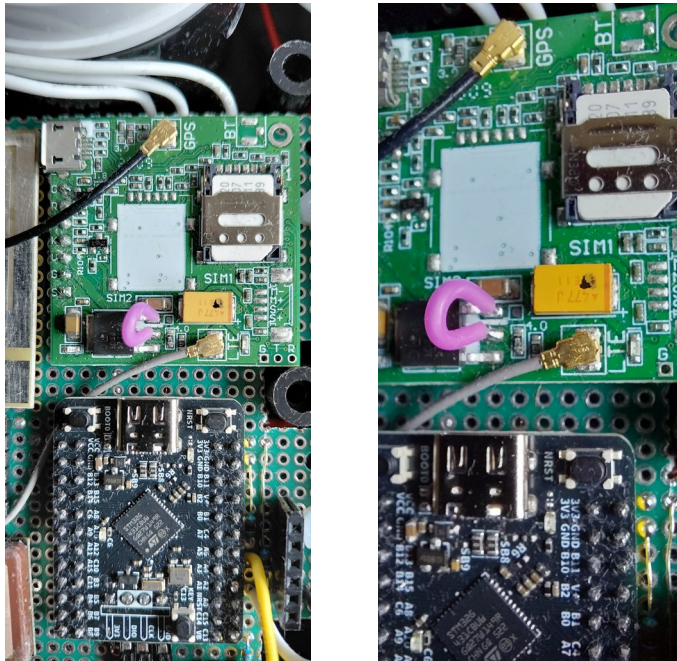
This project is powered by a 1S LiPo battery, with a supply voltage of around 3.7V. This is not a problem for the SAM M-10Q GNSS module and the STM32G431CB. The power supply **MUST** be connected to

SAM M-10Q	5V
STM32G431CB	VCC

When using an FTDI, do NOT connect the VCC of the FTDI, so as not to connect voltage in parallel to the LIPO

SIM A7670E

For the SIM A7670E, the voltage regulator must be bridged (see attached photos) and the supply voltage must be connected to V



When using a 1000 mAh LiPo, you will have a usage time of approximately 3 hours.

The SIM A7670E is set to a baud rate of 115200 baud upon delivery. This doesn't work reliably for me. Therefore, the program was written for a baud rate of 9600.

To set the SIM 7670E to 9600, connect a terminal, possibly the Arduino terminal, via an FTDI (set to 3.3V) to the SIM7670E.

If SIM A7670E is connected with LiPo do NOT connect FTDI VCC with module.

Set terminal baud rate to 115200

type

AT+IPREX=9600

module permanently sets the baud rate to 9600 baud.

Set terminal baud rate to 9600

type

AT

answer module

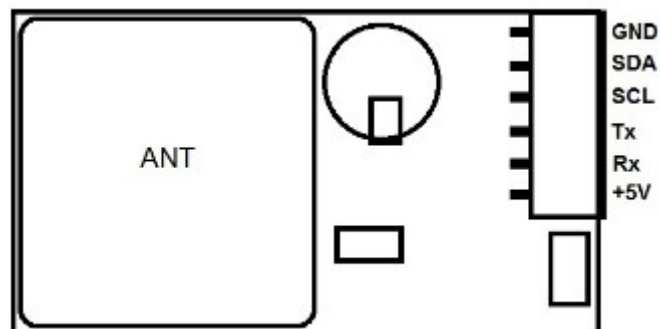
OK

SAM M10Q

!!! Connections SAM M-10Q !!!

This company uses an alternative color code for wiring.

DO NOT use the wire colors to connect the module, but use the following wiring diagram.



Connections

STM32G431CB

VCC			<<	VCC OUT TP4056
GND (2x)			<<	GND TP4056
PB10	M10Q_SERIAL	TX	>>	RX SAM-M10Q
PB11	M10Q_SERIAL	RX	>>	TX SAM-M10Q
PA2	A7670_SERIAL	TX	>>	R SIM A7670(E)
PA3	A7670_SERIAL	RX	>>	T SIM A7670(E)
PA0	ALARM INPUT			zie tekening
PA1	PRGM_EN			zie tekening
PA4	TEST			zie tekening
PA5	LEDS_1			zie tekening
PA6	LEDS_2			zie tekening
PA7	BUZZER			zie tekening

SIM A7670E

V			<<	VCC OUT TP4056 / LiPo na ON/OFF switch
G (2x)			<<	GND TP4056 / LiPo

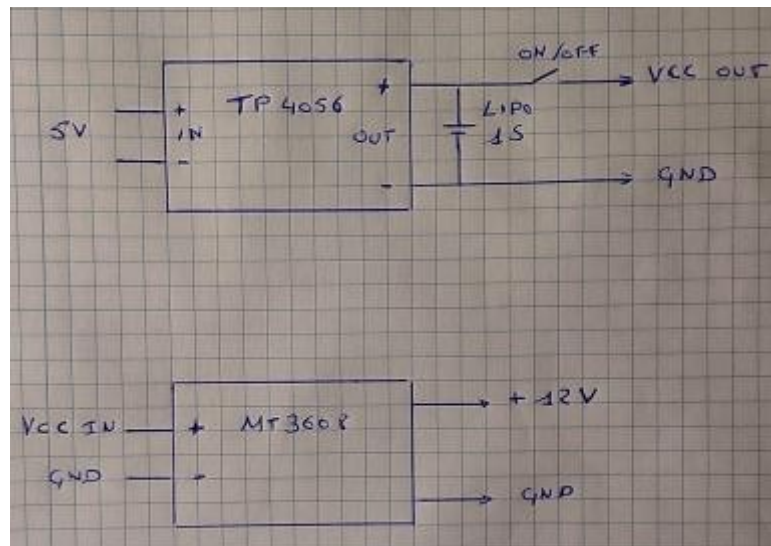
For supply voltage and bridging voltage regulator see previous pages

T	SIM A7670(E)	TX	<<	PA3 STM32G431CB
R	SIM A7670(E)	RX	<<	PA2 STM32G431CB

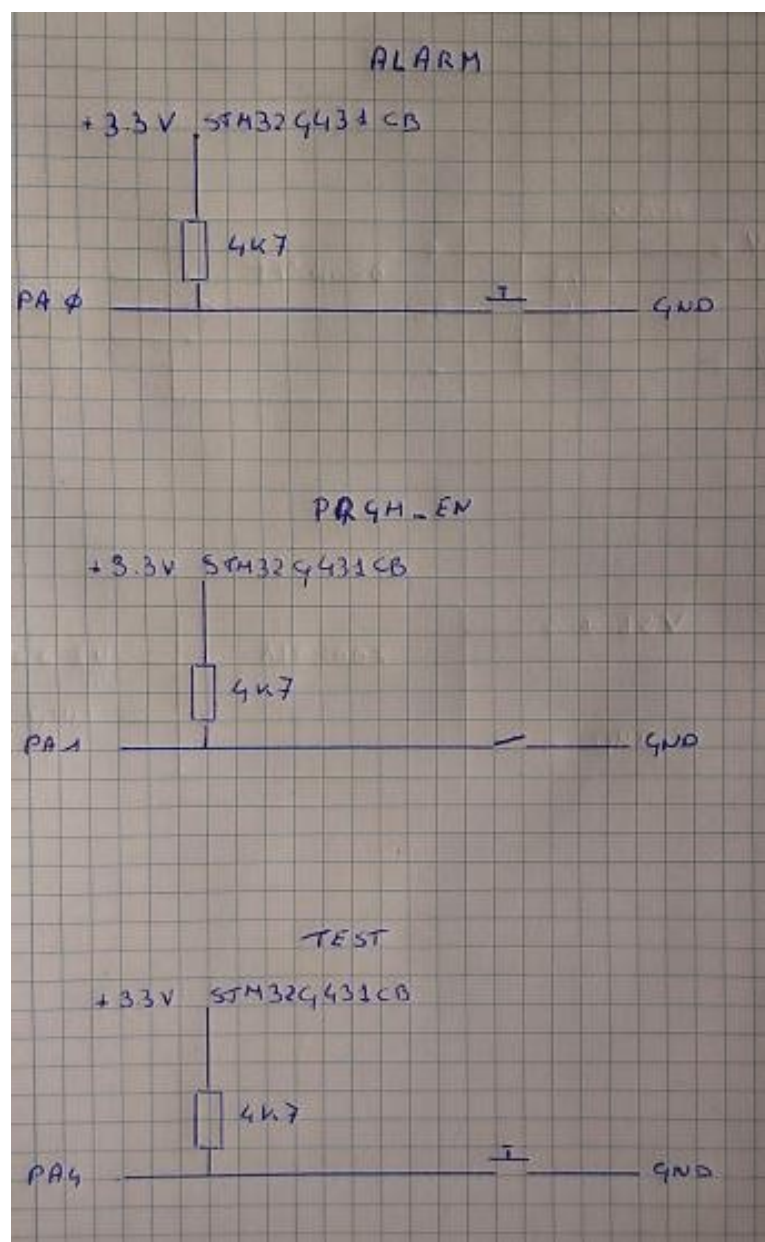
SAM M10Q

+5V			<<	VCC OUT TP4056 / LiPo na ON/OFF switch
GND			<<	GND TP4056 / LiPo
RX			<<	PB10 STM32G431CB
TX			<<	PB11 STM32G431CB

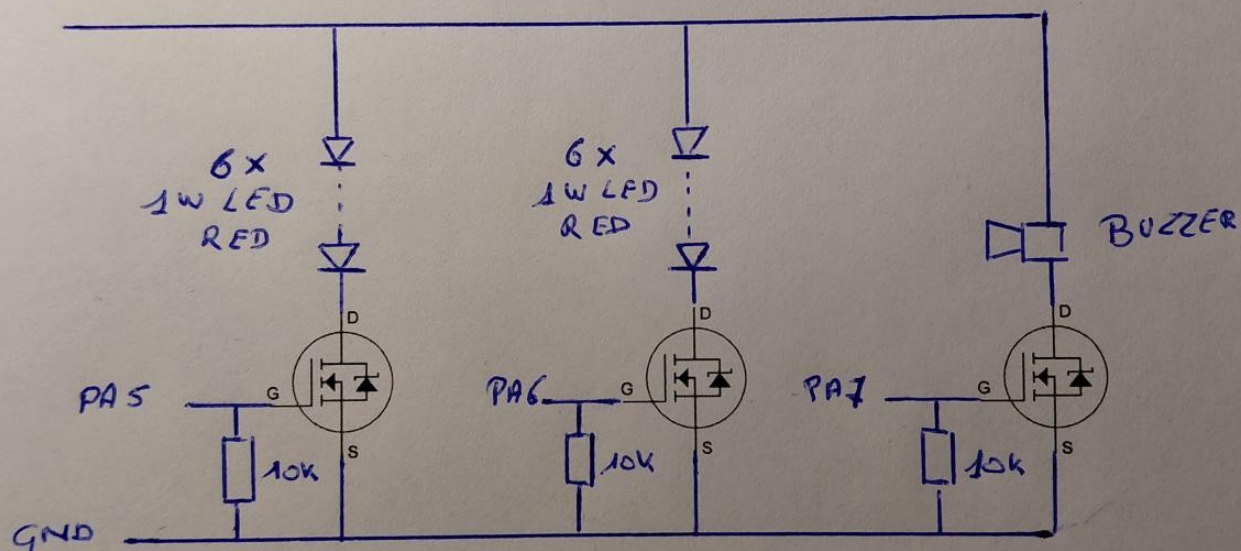
For connecting SAM M10Q see diagram on previous page



set MT3608 to 12V



+12V MT3608



3 x IRL540N

After building it's time to load the program.

Important:

Change the baud rate of the SIM A7670E to 9600, see page 7.

Download project

<https://github.com/thieu-b55/Personal-Alarm-Dual-GPS>

Open folder <Persoonlijk_Alarm_Dual_GPS> in Platform.io

To activate the buzzer in case of alarm, remove the // in line 145.

To avoid problems with incompatible updates, a .bin file is also available in the bin folder. The buzzer is enabled in this file.

The .bin file can easily be uploaded using the STM32CubeProgrammer.

<https://www.st.com/en/development-tools/stm32cubeprog.html>

This alarm system is built with two GNSS modules: one in the SIM7670E module and a separate SAM-M10Q module.

This isn't necessary, of course, and one can always be omitted if desired.

It's difficult to say which of the two will give the best result.

Project and software on Github

<https://github.com/thieu-b55/Personal-Alarm-Dual-GPS>

Good luck ,

thieu-b55

